

Coordinate Geometry
All expected concepts revision



MATHS HOTSHOT | SSC CGL 2026

by PARMAR SSC

Playlist • Public • 4 videos • No views

Play all

Date published (newest)

1 

SSC CGL 2026 | QUES. BASED ON SQUARE ROOT | MATHS HOTSHOT

PARMAR SSC • 64K views • 1 day ago

2 

SSC CGL 2026 | THESE DECIMAL & FRACTION CONCEPTS CAN TRICK YOU!

PARMAR SSC • 67K views • 2 days ago

3 

SSC CGL 2026 | MUST KNOW RULE FOR RATIONAL & IRRATIONAL NUMBERS

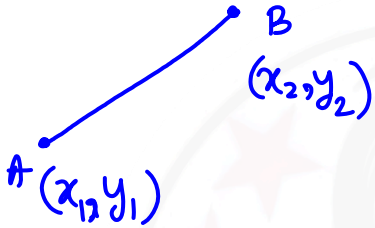
PARMAR SSC • 94K views • 2 days ago

4 

SSC CGL 2026 | MATHS HOTSHOT | NUMBER SYSTEM CLASSIFICATION

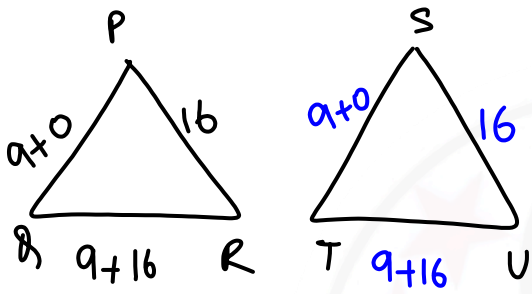
PARMAR SSC • 139K views • 3 days ago

Distance Formula (दूरी सूत्र):



$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$



Triangle PQR has vertices P(1,1), Q(4,1), and R(1,5). Triangle STU has vertices S(6,2), T(9,2), and U(6,6). Are these two triangles congruent?

त्रिभुज PQR के शीर्ष P(1,1), Q(4,1), और R(1,5) हैं। त्रिभुज STU के शीर्ष S(6,2), T(9,2), और U(6,6) हैं। क्या ये दोनों त्रिभुज सर्वांगसम हैं?

a) Yes, by SSS/हाँ, SSS द्वारा

b) Yes, by SAS/हाँ, SAS द्वारा

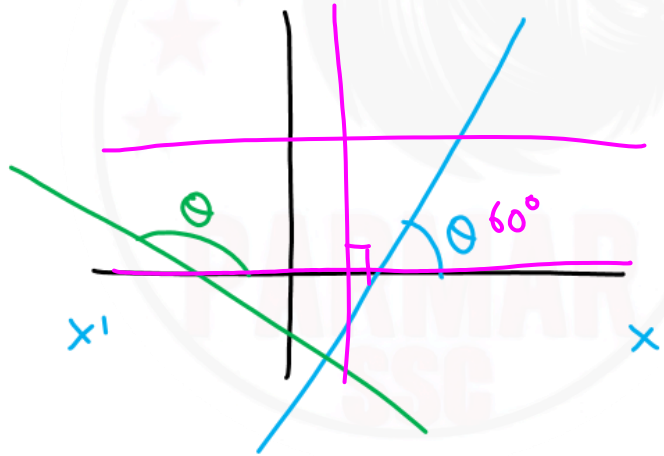
c) Yes, by ASA/ हाँ, ASA द्वारा

d) No, they are not congruent/नहीं, वे सर्वांगसम नहीं हैं

Slope/gradient of a line: (ढाल/प्रवणता/ढलान)

- **Slope of a line represents its degree of inclination.**

→ angle of line with +ve direction of x-axis.



i) $\text{Slope} = m = \tan\theta$

(θ = angle made by the line with the positive x-axis)

ii) $\text{Slope} = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{y_1 - y_2}{x_1 - x_2}$

((x_1, y_1) and (x_2, y_2) are two distinct points on the line)

iii) $ax + by + c = 0$

$\text{Slope} = -\dfrac{a}{b} = \dfrac{-(\text{Coeff. of } x)}{(\text{Coeff. of } y)}$

$2x + 3y + 5 = 0$

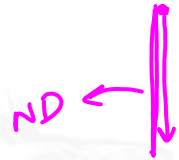
$\dfrac{-2}{3}$

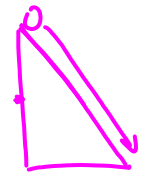
iv) $y = mx + c$

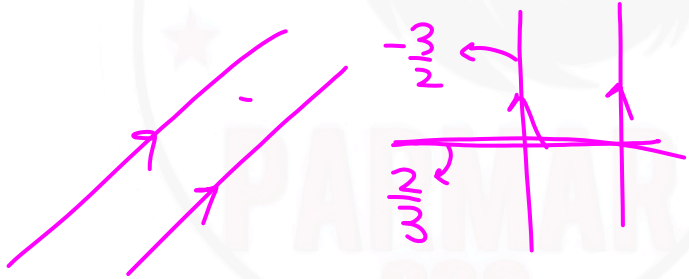
(m = slope of the line, c = y-intercept) $\rightarrow x = 0 \Rightarrow y = c$

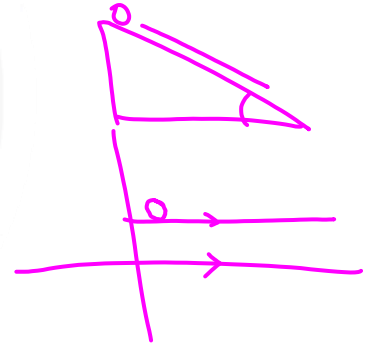
slope

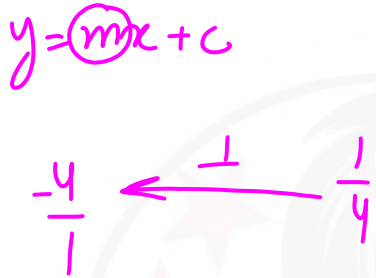
- **Slope of $x-axis$ is 0.**
- **Slope of $y-axis$ is not defined.**
- **Slopes of two parallel lines are same $i.e. m_1 = m_2$**
- **Slopes of two perpendicular lines are negative reciprocals of each other $i.e. m_1 = -\frac{1}{m_2}$ or $m_1 \times m_2 = -1$**











Find the slope of the line perpendicular to $y = \frac{1}{4}x + 7$.

$y = \frac{1}{4}x + 7$ पर लंबवत रेखा का ढलान ज्ञात कीजिए।

a) -4

b) 4

c) $-\frac{1}{4}$

d) $\frac{1}{4}$

PARMAR SSC

The lines $2x - 3y = 6$ and $kx + y = 4$ are perpendicular. What is the value of k?

रेखाएँ $2x - 3y = 6$ और $kx + y = 4$ लंबवत हैं। k का मान क्या है?

a) 1/2 b) 2/3

c) 2 d) 3/2



Equation of a line
(रेखा की समीकरण)

PARMAR SSC

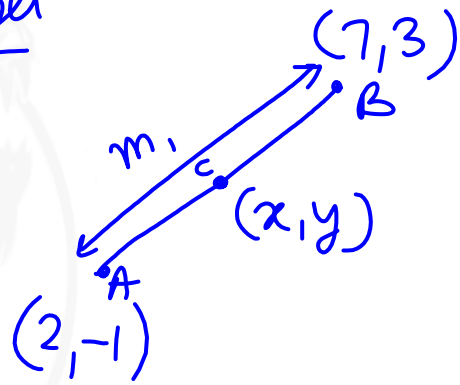
Two-point form:

$m_{AC} = m_{AB}$

$\frac{y+1}{x-2} = \frac{4}{5}$

$-4x+5y = \frac{4}{5}$

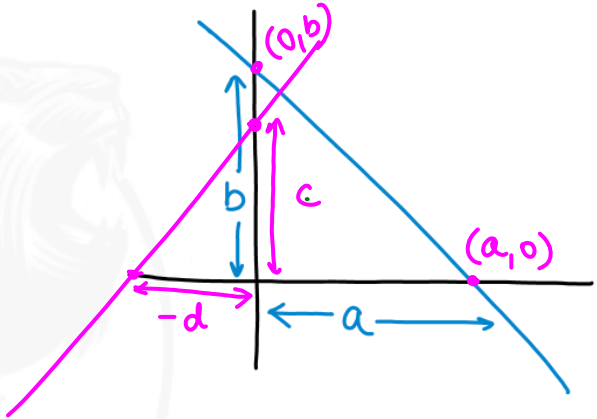
Example



Intercept form:

$$\frac{x}{a} + \frac{y}{b} = 1$$

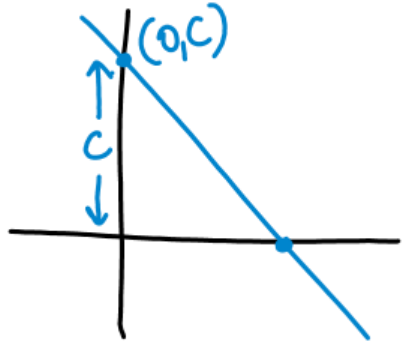
$$\frac{x}{-d} + \frac{y}{c} = 1$$



PARMAR SSC

Slope intercept form:

$y = mx + c$



Q. Eqn of line, if $m = -\frac{7}{2}$ & $c = 5$.

$y = -\frac{7}{2}x + 5$

PARMAR SSC

A line passes through the point (5, 7) and has a slope of –1. What is the equation of the line?

एक रेखा बिंदु (5, 7) से होकर गुजरती है और उसका ढाल –1 है। रेखा का समीकरण क्या है?

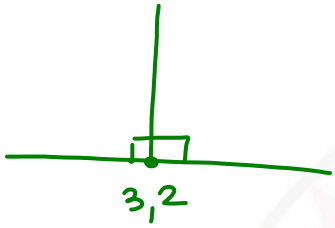
a) $y = -x + 12$

b) $y = \frac{x}{5} + 19$

c) $y = -2x - 4$

d) $y = 4x - 6$

PARMAR SSC



Two lines are perpendicular and intersect at the point (3,2). If one line has the equation y=2x-4, what is the equation of the other line?

दो रेखाएँ लंबरूप हैं और बिंदु (3,2) पर प्रतिच्छेद करती हैं। अगर एक रेखा का समीकरण y=2x-4 है, तो दूसरी रेखा का समीकरण क्या है?

a) y=-1/3x+7/2
b) y=-1/2x+7/2
c) y=-3/2x+7/2
d) y=-1/2x+5/2

$m = \frac{2}{1}$, $\frac{1}{-\frac{1}{2}}$

$-\frac{1}{2} \times 3 + \frac{7}{2}$

A line passes through the point (2, -4) and is parallel to the line $y = 4x + 1$. What is its equation?

एक रेखा बिंदु (2, -4) से होकर गुजरती है और रेखा $y = 4x + 1$ के समानांतर है। इसका समीकरण क्या है?

a) $y = 4x - 11$

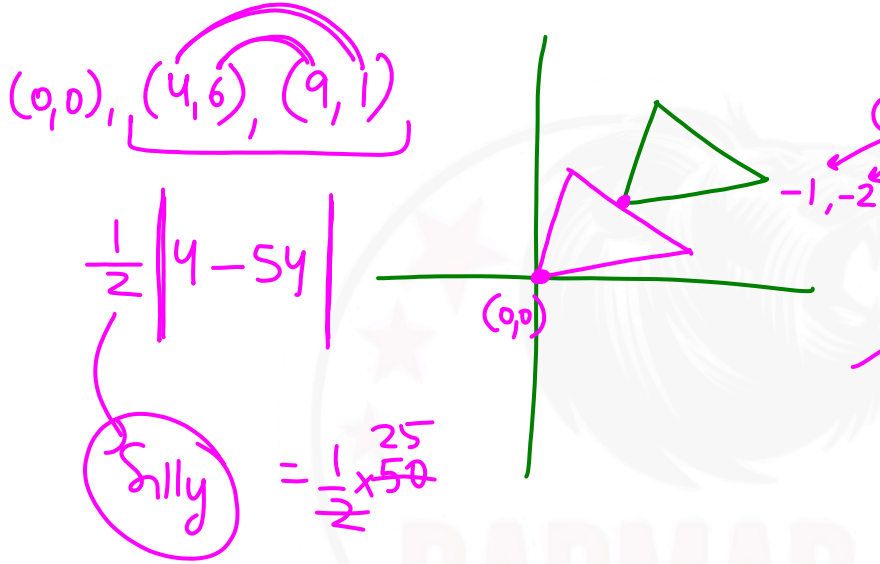
b) $y = 4x - 12$

c) $y = 4x - 10$

d) $y = 3x - 12$

Area of Triangle & Quad.

PARMAR SSC



What is the area of a triangle with vertices at (1,2), (5,8), and (10,3)?

(1,2), (5,8) और (10,3) शीर्षों वाले त्रिभुज का क्षेत्रफल क्या है?

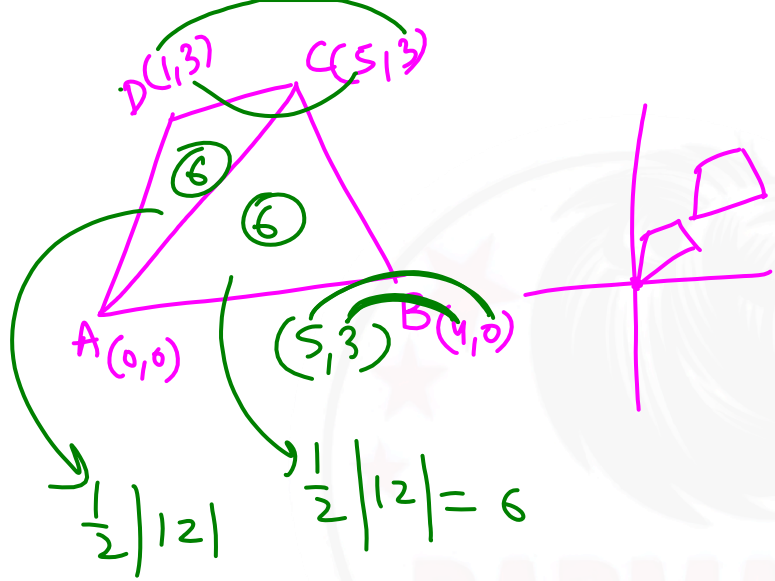
a) 10 square units

b) 15 square units

c) 20 square units

d) 25 square units

PARMAR SSC



A land plot is defined by points A(0,0), B(4,0), C(5,3), D(1,3). What is the area of the parallelogram representing this plot?

एक भूमि प्लॉट बिंदुओं A(0,0), B(4,0), C(5,3), D(1,3) से परिभाषित है। इस प्लॉट का क्षेत्रफल ज्ञात कीजिए जो एक समान्तर चतुर्भुज दर्शाता है।

a) 12
b) 8
c) 14
d) 10

Centroid (केन्द्रक):

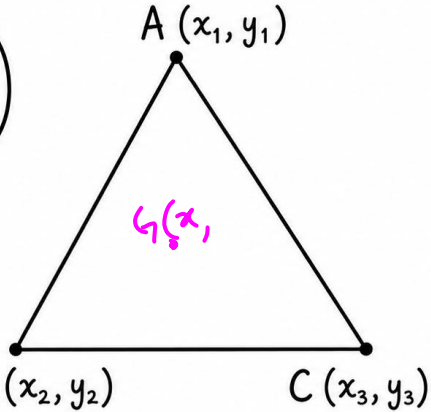
$$\text{Centroid} = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

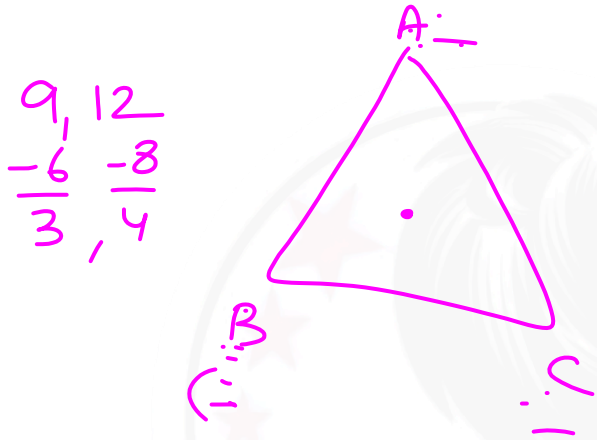
Avg. of x coordinates

Avg. of y coordinates

x_1, y_1 ——— x_2, y_2

$\left(\frac{x_1 + x_2}{2}, \right)$





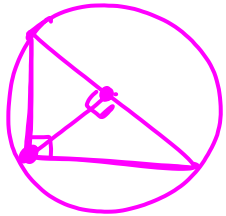
9, 12
−6 −8
3, 4

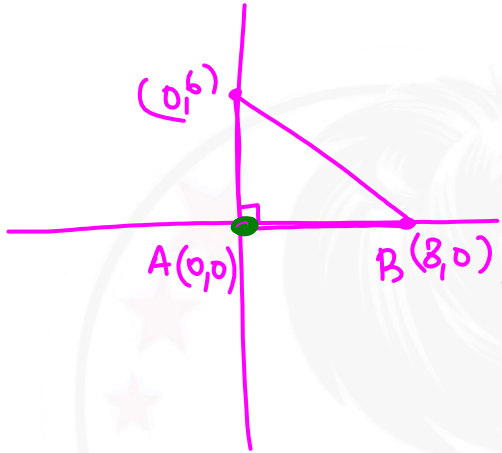
In triangle ABC, if the coordinates of the centroid are (3,4), and two vertices are (1,2) and (5,6), find the third vertex.

त्रिभुज ABC में, यदि केन्द्रक के निर्देशांक (3,4) हैं, और दो शीर्ष (1,2) और (5,6) हैं, तो तीसरा शीर्ष ज्ञात कीजिए।

a) (3,4)
b) (2,3)
c) (4,5)
d) (3,5)

PARMAR SSC

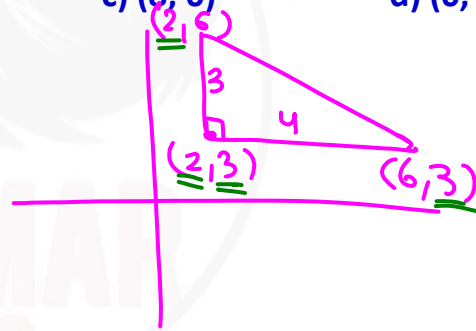


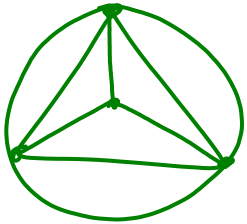


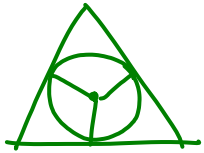
The coordinates of the vertices of a triangle are A(0,0), B(8,0), and C(0,6). What is the location of the orthocenter?

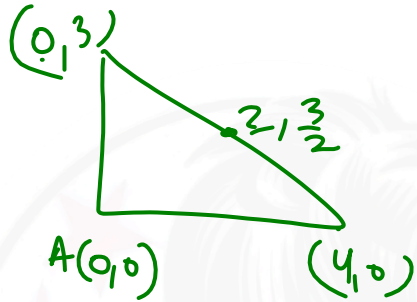
एक त्रिभुज के शीर्षों के निर्देशांक A(0,0), B(8,0) और C(0,6) हैं। लम्बकेन्द्र का स्थान क्या है?

a) (0, 0)
b) (4, 3)
c) (8, 0)
d) (0, 6)









Point P is at an equal distance from the three vertices of a triangle with coordinates A(0,0), B(4,0), and C(0,3). What are the coordinates of point P?

बिंदु P, निर्देशांक A(0,0), B(4,0), और C(0,3) वाले त्रिभुज के तीनों शीर्षों से समान दूरी पर है। बिंदु P के निर्देशांक क्या हैं?

a) (2, 1.5)
b) (2, 2)
c) (1.5, 2)
d) (2.5, 2.5)

PARMAR SSC

For full & deep revision →
- FRB (2x)
- FRB concept notes
- FRB annotated pdf

PARMAR SSC